Eco-Friendly Daily Commute Helper: A Quantitative, Scenario-Based Study, Simulated Analytical Demonstration, and AI-Driven Conceptual Framework for Sustainable Urban Commuting

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Research-integrity statement. This paper reports a *proposed* quantitative research design and a *simulated* analytical demonstration. No survey was administered, no participant was recruited, and no system was deployed. Every numerical result reported in this paper was generated from a synthetic dataset produced by a computer program with a fixed random seed. These numbers describe no real person and no real system, and they must not be cited as empirical evidence about Chandigarh University students or about any deployed commute assistant.

Abstract

Background and problem. Daily commuting is a substantial contributor to urban greenhouse-gas emissions, and commuter mode choice is an important point of intervention in climate mitigation (Eslami, 2025; Creutzig et al., 2015). Sustainable alternatives exist—public transport, active travel, micromobility, shared mobility and electric vehicles—but the literature tends to examine these modes in isolation, and integrated, personalised, behaviourally aware commute assistants remain under-researched (Kashem et al., 2025; Ouyang et al., 2025). **Research gap.** The retained literature does not provide a measurable design that links commuter-level awareness, attitudes and practical priorities to responses to a personalised recommendation, nor a transparent demonstration of how such a design would be analysed and connected to an emission model. **Aim and design.** This paper specifies a quantitative, cross-sectional, scenario-based research design with a conceptual system-evaluation component. It states six research questions and six testable hypotheses, defines measurable constructs and a seven-section instrument (including six scenario-based commute-choice tasks and a baseline-versus-recommendation task), specifies an emission model, and sets out a statistical analysis plan for a proposed convenience sample of undergraduate students at Chandigarh University (target $n \approx 150$). **Simulated demonstration.** No genuine survey has been conducted. To demonstrate the analytical procedure end to end, a synthetic dataset of $N = 120$ respondents was generated with a fixed random seed. In the simulated data, mean sustainable-mobility awareness was 3.52 ($SD = 0.76$) and mean sustainable travel intention 3.48 ($SD = 0.72$) on 1–5 scales; awareness and intention were positively associated ($\rho = .34$, $p < .001$). A binary logistic model of baseline low-emission mode selection was

significant overall ($\chi^2(6) = 35.52$, $p < .001$; McFadden's pseudo-$R^2 = .23$; AUC = .83), with cost sensitivity a positive predictor and perceived convenience a negative predictor. In the simulated baseline-versus-recommendation task across 720 paired decisions, the private-car share fell from 35.0% to 25.6% and modelled emissions fell from 717.0 to 582.8 g CO_2 per trip (18.7% reduction). These are model-based results, partly built into the data generator, and are *not* evidence of real behavioural or environmental effects. **Framework and limitations.** The original five-layer architecture (input; data acquisition; AI processing; decision and behavioural nudging; feedback and evaluation) is retained and linked explicitly to the research variables and evaluation metrics. Principal limitations are the simulated data, the proposed (not completed) sampling, self-report and scenario-based measurement, uncertain emission factors, and the absence of any real deployment. The framework is conceptual and has not been implemented or empirically validated.

Keywords: Eco-friendly commuting; sustainable urban mobility; multimodal transport; micromobility; Mobility-as-a-Service; behavioural nudging; artificial intelligence; smart commute assistant; simulated data; carbon emissions.

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1 Introduction

1.1 Background

Daily commuting demand has grown with urbanisation and economic growth, and transport systems worldwide have become heavily relied upon as a result (Lättman & Otsuka, 2024). The transport sector is a significant source of greenhouse-gas emissions, energy use and environmental degradation, which makes sustainable mobility a central concern in research and policy (Collazos et al., 2024; Ndhlovu et al., 2025). Daily commuting accounts for a significant share of a city's carbon footprint and is therefore a major focus of climate mitigation (Eslami, 2025); transport more broadly has been described as a roadblock to climate mitigation (Creutzig et al., 2015).

Eco-friendly commuting can be defined as the use of transport modes that offer reduced emissions, lower energy consumption and more responsible mobility (Lättman & Otsuka, 2024). It can involve public transport, electric vehicles, walking and cycling, shared mobility and micromobility such as e-bikes and e-scooters (Comi & Polimeni, 2024; Ndhlovu et al., 2025). Micromobility is considered promising for replacing short journeys made by personal vehicle, improving accessibility, reducing congestion and lowering pollutant emissions (Comi & Polimeni, 2024), and shared mobility applications and systems can reduce personal car use and the overall carbon footprint (Garus et al., 2024). Proximity to services, active mobility and urban design also matter for commuter-related emissions (Marzolla et al., 2025).

Artificial intelligence, data analytics and digital mobility platforms make it possible to build smart commute helpers that direct users towards environmentally friendly travel choices. Behavioural incentive schemes such as eco-driving suggestions and reward-based digital platforms have been reported to reduce emissions by promoting sustainable commuting behaviour (Niazi et al., 2024). Electric vehicles are widely promoted as a route to decarbonising transport, although their suitability depends on energy supply, infrastructure availability and user uptake (Ndhlovu et al., 2025).

1.2 Problem Statement

Despite the availability of eco-friendly commuting options, several obstacles impede their adoption, including inadequate infrastructure, behavioural barriers, economic factors and policy restrictions (Collazos et al., 2024). The environmental benefit of electrified options is itself conditional on energy supply and infrastructure (Ndhlovu et al., 2025). Interest is therefore growing in commuting assistants that combine technology with behavioural consideration and guide people towards environmentally responsible options (Eslami, 2025). Such systems aim not only to reduce emissions but also to improve liveability, health and transport efficiency, which requires informed decision-making and personalised recommendations (Lättman & Otsuka, 2024).

The problem addressed here has two parts. First, it is not clear how commuters' awareness, attitudes, practical priorities (cost, time, convenience) and receptiveness to digital recommendations jointly shape the choice of low-emission modes. Second, there is no measurable evaluation design that links such commuter-level variables to the components of an AI-driven commute assistant and to modelled emission outcomes.

1.3 Research Gap

Although interest in eco-friendly commuting has grown, research tends to focus on specific transport modes rather than on integrated commuting ecosystems that combine several sustainable options (Kashem et al., 2025). Personalised digital commute assistants that integrate environmental-impact data, behavioural nudges and real-time transport information are underresearched (Ouyang et al., 2025), as are interdisciplinary approaches that bridge urban planning,

behavioural science and digital technology (Eslami, 2025). The original version of this paper responded with a conceptual framework only. The gap that remains, and that this revision addresses, is the absence of (i) a measurable, quantitative research design linking commuter variables to recommendation response, and (ii) a transparent, reproducible demonstration of how such a design would be analysed and connected to an emission model and to the proposed architecture.

1.4 Aim

The aim of this paper is to design a quantitative, scenario-based research study of sustainable commuting awareness, intention, mode choice and response to personalised recommendations; to demonstrate its analysis procedure using a clearly labelled simulated dataset; and to connect the design to an AI-driven, behaviourally aware *Eco-Friendly Daily Commute Helper* with a model-based environmental evaluation.

1.5 Objectives

O1. To review sustainable commuting modes (active, public, electric/low-emission, micromobility and shared) and their emission-reduction potential (Section 2).

O2. To examine the digital technologies (AI, IoT, big data and Mobility-as-a-Service) that support commuting decisions (Section 2).

O3. To examine behavioural mechanisms (feedback, nudging, gamification and social comparison) that may shift commuters towards low-emission modes (Section 2).

O4. To specify how awareness, environmental concern and sustainable travel intention would be measured, and to demonstrate their analysis using simulated data (Sections 3–9).

O5. To examine, in the simulated data, factors (awareness, concern, convenience, cost sensitivity, time sensitivity and assistant acceptance) associated with selecting a low-emission mode.

O6. To evaluate, in a simulated scenario task, response to personalised recommendations (recommendation acceptance and mode shift) relative to baseline choice.

O7. To estimate model-based CO_2 savings per trip and illustrative monthly and annual projections from the simulated data (Sections 7 and 9).

O8. To develop the five-layer framework and map research variables to system inputs, AI processes, behavioural interventions and evaluation metrics (Sections 10 and 11).

1.6 Research Questions

RQ1

What is the level of sustainable-mobility awareness and environmental concern among (proposed) university commuters?

RQ2

What is the level of sustainable travel intention, and does it differ between private-vehicle users and other commuters?

RQ3

Which factors (awareness, concern, convenience, cost sensitivity, time sensitivity and assistant acceptance) are associated with choosing a low-emission mode at baseline?

RQ4
Are environmental awareness and environmental concern associated with sustainable travel intention and with mode shift after a recommendation?

RQ5
How acceptable are personalised recommendations, and is perceived convenience associated with acceptance?

RQ6
How do mode selection and modelled CO_2 emissions differ between baseline choice and post-recommendation choice?

1.7 Hypotheses

The following testable hypotheses are specified for the proposed study. Each is tested on the simulated data in Section 9 solely to demonstrate the analysis; outcomes are not findings about real commuters.

H1 Environmental awareness is positively associated with sustainable travel intention.

H2 Environmental concern is positively associated with the number of car-to-sustainable mode shifts after a recommendation.

H3 Perceived convenience is positively associated with acceptance of sustainable commute recommendations.

H4 Private-vehicle users report lower sustainable travel intention than other commuters.

H5 The six predictors (awareness, concern, convenience, cost sensitivity, time sensitivity and assistant acceptance) jointly predict selection of a low-emission mode at baseline.

H6a
Personalised recommendations reduce the private-car share of simulated decisions.

H6b
Personalised recommendations reduce modelled CO_2 per trip.

1.8 Significance of the Study

The study offers an integrated view of a field in which modes are often examined in isolation and interdisciplinary approaches remain under-explored (Eslami, 2025; Kashem et al., 2025). It supplies a measurable research design, instrument, data dictionary, emission model and analysis plan that future researchers can reuse with genuine data, and a framework in which every research variable has a defined place in a proposed commute assistant (Liu et al., 2025). Because eco-friendly commuting can also improve urban air quality, health and the efficiency of urban movement (Cui & Zhang, 2024), the design has applied as well as methodological value. Methodologically, the paper demonstrates a transparent practice for reporting simulated results alongside a proposed empirical study.

2 Literature Review

Sustainability in transportation has emerged as a key research theme because the sector contributes significantly to carbon emissions and environmental deterioration (Eslami, 2025; Ouyang et al., 2025). Commuting behaviour is a major source of urban greenhouse-gas emissions, and

daily travel behaviour is one of the most important intervention points in climate-mitigation strategies (Eslami, 2025). Sustainable urban mobility integrates environmental science, transportation engineering, behavioural studies and digital technologies (Ouyang et al., 2025). The review below retains the 24-source base of the original paper and is organised by the mode, technology and behaviour themes that structure the proposed measurement design. Statements in this section are literature claims; interpretation is flagged as such.

2.1 Sustainable Urban Mobility

Urban mobility studies stress the need to reduce personal-vehicle use in order to lower emissions, congestion and energy consumption (Comi & Polimeni, 2024). Large-scale comparisons of urban spatial planning indicate that proximity-based city models, such as the 15-minute city, can substantially reduce transport-related CO_2 by encouraging walking and cycling, which implies that sustainable commuting depends strongly on infrastructure design and urban planning (Marzolla et al., 2025). *Interpretation:* because commute distance and local infrastructure condition the feasible options, the proposed instrument records a commute-distance category and the scenarios vary distance.

2.2 Active Transportation

Walking and cycling emit almost no pollution and have health-related benefits (Lättman & Otsuka, 2024). Improved walking infrastructure and dedicated cycling lanes contribute substantially to active commuting (Marzolla et al., 2025). Active modes therefore appear as the bicycle option in the short- and medium-distance scenarios (Appendix B) and as the walking and cycling intention items.

2.3 Public Transportation

Buses, metro systems and trains can be more energy-efficient than privately owned vehicles because they carry many people at once, and the accessibility and reliability of public transport can reduce congestion and emissions. Real-time transit information and integrated ticketing improve usability and sustainability (Cui & Zhang, 2024). Cats et al. (2017) examined the prospects of fare-free public transport using evidence from Tallinn, a reminder that pricing is one lever on public-transport use; this paper draws no finding from that study beyond its stated subject.

2.4 Electric and Low-Emission Mobility

Electric vehicles have become a significant technology for lowering transport emissions, especially when powered by renewable energy (Liu et al., 2025), and electrification is widely regarded as a key route to reducing transport emissions (Mogire, 2025). Their benefit is conditional on electricity generation sources, battery sustainability, infrastructure and user uptake (Ndhlovu et al., 2025; Ouyang et al., 2025). Organisational and policy measures together with technological innovation also contribute to energy conservation and emission reduction in road transport (Liu et al., 2025).

2.5 Shared Mobility and Carpooling

Carpooling, ride-sharing and Mobility-as-a-Service models can reduce the number of vehicles on the road and improve transport efficiency, and shared mobility has been proposed to lower personal carbon footprints (Cui & Zhang, 2024; Garus et al., 2024). Shaheen & Cohen (2018)

address shared-mobility policy in a California policy brief and are cited here only as earlier policy context.

2.6 Micromobility

Micromobility options, including bicycles, e-bikes and scooters, have been identified as a means of replacing short urban journeys with positive environmental and health effects (Comi & Polimeni, 2024). Light electric vehicles such as e-bikes and e-scooters can be efficient for short-distance commuting and for first- and last-mile connectivity (Mogire, 2025). Shared micromobility can complement public transport and help address first- and last-mile connectivity (Cui & Zhang, 2024), while the use of electric micromobility is influenced by demographics, time concerns, infrastructure accessibility and affordability (Wu et al., 2025). These conditions motivate the time-sensitivity, cost-sensitivity and accessibility variables in this study.

2.7 AI and Intelligent Transportation

Artificial intelligence, IoT and blockchain technologies are increasingly used to streamline transport performance and reduce environmental effects, enabling data-driven decision-making, traffic forecasting and intelligent mobility planning (Kashem et al., 2025). AI-driven transportation research also reports big-data analytics for emission forecasting, route optimisation and sustainable mobility planning (Ouyang et al., 2025). Organisational management and technological innovation through AI have been discussed as strategies for sustainable road transport (Liu et al., 2025).

2.8 IoT, Big Data and Real-Time Mobility Information

Research on carbon-neutral mobility hubs shows how AI can help optimise location choice, reduce vehicle distance travelled and improve integration with public transport; IoT connectivity and small electric vehicles are becoming increasingly important in shifting urban transport towards sustainability (Kashem et al., 2025; Mogire, 2025). These results inform the data-acquisition and AI-processing layers of the proposed assistant and can inform smart green commuting assistants that give individual green travel advice (Kashem et al., 2025).

2.9 Behavioural Nudging and Sustainable Travel

Behavioural change, including environmental conscientiousness and willingness to travel sustainably, is a determinant of success in eco-friendly commuting (Eslami, 2025). Psycho-social determinants of pro-environmental behaviour have long been synthesised through meta-analysis (Bamberg & Möser, 2007), which underpins the use of awareness, concern and intention constructs here. Combining environmental-impact monitoring with behavioural incentive programmes has been reported to improve acceptance of environmentally friendly commuting (Eslami, 2025). Social-norm messaging has been linked to reduced household energy consumption in extreme heat in the United Arab Emirates (Nisa et al., 2025); that context is energy use rather than commuting, so it is cited as indicative of message-based influence only.

2.10 Carbon Feedback and Gamification

Behavioural incentive schemes such as eco-driving suggestions and reward-based digital platforms have been shown to reduce emissions by promoting sustainable behaviour (Niazi et al., 2024). Sunio (2018) describes a mobility behaviour-change support system for sustainable campus commuting, which is closely related to the proposed setting. The original paper reported that

immediate feedback can strengthen pro-environmental behaviour (Nisa et al., 2025; Sunio, 2018). The specific effect sizes of feedback, gamified rewards and social comparison on commuting are not established in the retained source set, and are therefore treated as proposals for empirical testing.

2.11 Personalised Mobility and Mobility-as-a-Service

In MaaS, multimodal travel choices are coordinated through digital tools. Hensher (2024) discusses the future of MaaS, and a Sydney trial reports insights from MaaS users (Smith et al., 2023). Systematic reviews indicate that MaaS adoption does not necessarily have a positive environmental impact unless platform design is strongly oriented towards low-carbon travel options (An & Shen, 2025). Smart commute applications can offer real-time traffic information, estimated emissions and personalised route recommendations (Kashem et al., 2025; Ouyang et al., 2025), and combining impact monitoring with incentives has been associated with greater acceptance of eco-friendly commuting (Eslami, 2025). *Interpretation:* an assistant's design objective matters as much as its functionality, which is why the proposed ranking in Layer 4 includes CO_2 explicitly.

2.12 Research Gap

Three gaps follow from the review. First, research often concentrates on specific transport modes rather than integrated commuting ecosystems (Kashem et al., 2025). Second, personalised assistants that integrate environmental-impact data, behavioural nudges and real-time transport information are under-researched (Ouyang et al., 2025). Third, interdisciplinary work bridging urban planning, behavioural science and digital technology is limited (Eslami, 2025). To these, this revision adds a methodological gap: the retained sources do not provide a measurement design that links commuter-level awareness, attitudes and practical priorities to response to a personalised recommendation and to a transparent emission calculation.

2.13 Summary of Literature

Across the reviewed studies there is broad agreement that commuting is a significant source of urban emissions and that reducing reliance on private vehicles is central. Evidence on individual solutions is mostly positive but conditional: proximity-based planning and active mobility depend on infrastructure design (Marzolla et al., 2025); electric micromobility depends on demographics, time, infrastructure and affordability (Wu et al., 2025); and electrification depends on energy supply, infrastructure and battery technology (Ndhlovu et al., 2025; Ouyang et al., 2025). AI, IoT and big data are reported to support emission forecasting, route optimisation and personalised advice (Kashem et al., 2025; Ouyang et al., 2025). What is missing is an integrated, measurable treatment of these elements within a personalised commute assistant. Table 1 compares the principal studies; "Not stated" marks information that the source material available for this paper does not report and that was deliberately not inferred.

Table 1: Comparison of principal studies (from the 24-source base; "Not stated" = not reported in the source material used).

Study	Context	Method	Key focus	Main finding	Limitation	Relevance
Comi & Polimeni (2024)	Urban micromobility	Data-driven approach	Sustainability benefits of micromobility	Micromobility can replace short personal-vehicle journeys; benefits include accessibility, less congestion and lower pollutants	Not stated	Supports bicycle and e-bike options and distance categories
Marzolla et al. (2025)	Urban spatial planning, large-scale comparison (preprint)	Comparison of city models	Proximity-based cities and mobility CO_2	Proximity models (e.g., 15-minute city) can substantially reduce transport CO_2 via walking and cycling	Outcomes depend on infrastructure and urban design; preprint	Motivates commute-distance variable
Lättman & Otsuka (2024)	Urban active travel and public transport	Not stated	Sustainable urban mobility	Walking and cycling emit almost no pollution and offer health benefits; definition of eco-friendly commuting	Not stated	Definition and active-mode rationale
Cui & Zhang (2024)	Shared micromobility and public transit incl. grey literature	Systematic literature review	Integration of micromobility into transit	Combining micromobility with public transport helps first/last-mile connectivity	Not stated	Multimodal scenario (metro + e-bike)
Wu et al. (2025)	Electric micromobility mode choice (preprint)	Stated-preference, latent class choice	Determinants of e-micromobility use	Use influenced by demographics, time concerns, infrastructure accessibility and affordability	Preprint	Time and cost sensitivity variables
Garus et al. (2024)	Shared mobility services	Not stated	Environmental impact of shared mobility	Shared mobility can reduce personal car use and carbon footprint	Not stated	Carpool and shared-ride options
Ndhlovu et al. (2025)	Urban decarbonisation: EVs, public transport, infrastructure	Overview	Pathways to net-zero urban transport	EV suitability depends on energy supply, infrastructure and user uptake	Conditional benefits	Emission-factor caveats (Section 7)
Kashem et al. (2025)	Digitalised transport operations	Systematic review of AI, IoT, blockchain	Digitalisation for sustainable mobility	Digital tools support data-driven decisions, forecasting and planning; integrated ecosystems understudied	Not stated	Layers 2–3; research gap
Ouyang et al. (2025)	Transport carbon emissions	Review of AI and big-data studies	Emission forecasting and route optimisation	AI/big data support forecasting and optimisation; personalised assistants underresearched	Infrastructure, battery and policy gaps noted	Layer 3; research gap
Liu et al. (2025)	Road transport strategies	Technology and organisational management through AI	Not stated	Organisational and policy measures with technology contribute to energy saving and emission reduction	Not stated	Assistant design context
An & Shen (2025)	Mobility-as-a-Service	Systematic literature review	MaaS and sustainable travel behaviour	MaaS adoption is not automatically environmentally positive unless design is low-carbon oriented	Not stated	CO_2-aware ranking in Layer 4

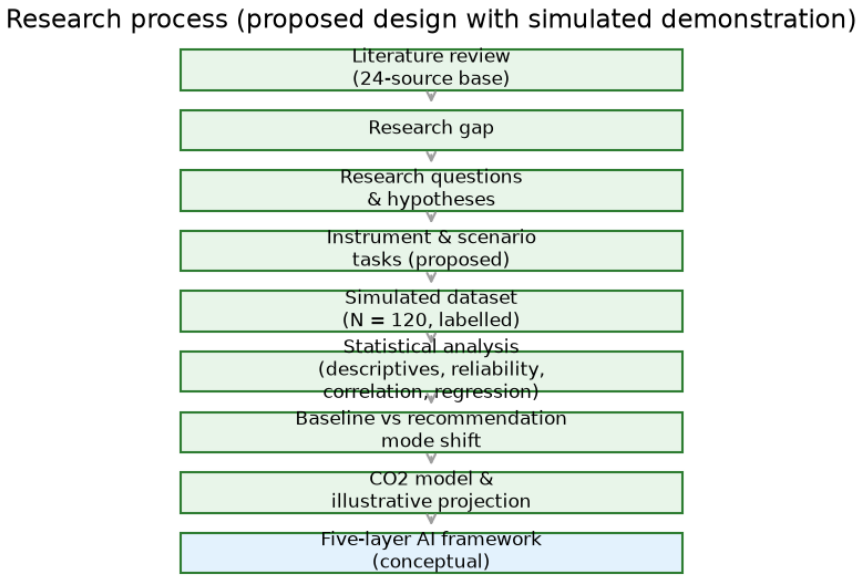


Figure 1: Research process: literature and gap lead to a proposed design and instrument, a labelled simulated dataset, a full statistical demonstration, a scenario experiment and an emission projection, and finally to the conceptual five-layer AI framework.

3 Research Methodology

This study uses a **quantitative, cross-sectional, scenario-based research design with a conceptual system-evaluation component**. The empirical version of the study would collect responses from undergraduate students; in this paper, a simulated dataset is used to demonstrate the complete analytical procedure. Throughout, the instrument, sampling and analysis are described as *proposed*, and every numerical result is described as a *simulated demonstration*.

3.1 Research Design

The design is quantitative because it measures constructs numerically and tests hypotheses with statistical procedures; cross-sectional because it observes respondents at a single point in time; and scenario-based because mode choice is elicited through structured hypothetical trips rather than observed travel. A within-subject baseline-versus-recommendation task adds a quasi-experimental element, although it is hypothetical and therefore cannot establish causal effects. The conceptual system-evaluation component maps each measured construct to a component of the proposed five-layer assistant (Section 10).

3.2 Research Setting

The proposed setting is the Gharuan campus of Chandigarh University, Punjab, India. The campus attracts commuters from surrounding urban and semi-urban areas, which produces variation in commute distance and available modes. The setting is a single institution, which limits generalisability (Section 13).

3.3 Target Population

The target population is undergraduate students who travel to campus at least three days per week. Students who study fully online or who live on campus with no commuting trip are outside

the population of interest.

3.4 Sampling Technique

A convenience (non-probability) sample is proposed, recruited through departmental mailing lists and class announcements. Convenience sampling is used because no sampling frame of commuting students is available; the resulting selection bias is acknowledged as a limitation.

3.5 Proposed Sample Size

The target is $n \approx 150$ complete responses. This is adequate for descriptive analysis, reliability estimation and the planned group comparisons, and is near the upper end of the common 10-observations-per-item guideline for factor analysis of a 40-item instrument; it is modest for the logistic model and would constrain the number of predictors (Section 13). No power calculation is claimed, because no pilot variance estimate exists yet.

3.6 Inclusion and Exclusion Criteria

Inclusion: age 18 or older; currently enrolled undergraduate; commutes to campus at least three days per week; provides informed consent. *Exclusion:* failed attention check; completion time below one-third of the median (speeding); straight-lining across a Likert block.

3.7 Research Instrument

A self-administered online questionnaire with seven sections (A–G) is proposed. Section A records demographics and the commuting profile; Sections B and C measure sustainable-mobility awareness and environmental concern; Section D measures commuting decision factors; Section E measures sustainable travel intention; Section F presents six scenario-based commute-choice tasks; and Section G presents personalised recommendation cards and measures assistant acceptance. The full instrument is in Appendix A.

3.8 Questionnaire Structure

Table 2 summarises the structure. The instrument uses five-point Likert items and single-choice scenario items. One unscored attention-check item is embedded in Section B.

Table 2: Proposed questionnaire structure.

Section	Content	Items	Scale
A	Demographics and commuting profile	7	Categorical / count
B	Sustainable-mobility awareness	9 (+1 attention)	1–5 agreement
C	Environmental concern and attitudes	7	1–5 agreement
D	Commuting decision factors	12	1–5 agreement
E	Sustainable travel intention	7	1–5 willingness
F	Scenario-based commute choice (Stage A)	6	Single choice
G1	Recommendation response (Stage B/C)	6	Single choice
G2	Assistant acceptance	7	1–5 agreement

3.9 Scenario-Based Commute Choice Task

Six scenarios (S1–S6) vary distance (3–15 km), context (weather, peak hour, first/last mile, personal safety) and the option set (private car, city bus, metro/rail, bicycle, e-bike, carpool, shared ride, and one multimodal metro+e-bike option). In Stage A, respondents see travel time, cost, comfort and convenience but *not* estimated CO_2; they choose their preferred option. In Stage B they see a personalised recommendation card that displays travel time, cost, CO_2 impact, convenience and an environmental-benefit statement. In Stage C they choose again. Withholding CO_2 in Stage A allows the effect of carbon information to be isolated. Scenario values are illustrative design values, not measurements of real services in the Chandigarh region (Appendix B).

3.10 Variables and Operational Definitions

Variables, scales and coding are defined in Section 5 and the data dictionary in Section 6.

3.11 Scoring Procedure

Scale scores are the arithmetic mean of their items (range 1–5), so that scores are comparable across scales of different length. Mode choices are coded to categories; a *sustainable* (low-emission) choice is any option other than the solo private car. Emission variables use the model in Section 7. Full coding rules are in Appendix C.

3.12 Pilot Testing

A pilot with approximately 15–20 students is proposed to assess wording clarity, completion time, scenario realism and technical issues, and to obtain preliminary reliability estimates. Items with a corrected item–total correlation below about .30, or scales with a preliminary alpha well below .70, would be reviewed. Pilot data would be merged with main data only if the instrument is unchanged.

3.13 Validity

Content validity is supported by grounding items in the reviewed literature (Section 2) and by review from the supervising faculty member and, ideally, two additional subject experts. Face validity is assessed in the pilot. Construct validity would be examined with exploratory factor analysis (if sample size permits), convergent patterns among related scales and known-group differences (for example, private-vehicle users versus others). Criterion validity against observed travel behaviour is not available in the present design.

3.14 Reliability

Internal consistency would be assessed with Cronbach's alpha for each multi-item scale (conventional benchmark $\geq .70$, with lower values interpreted cautiously for short three-item scales) together with corrected item–total correlations and alpha-if-item-deleted. The alpha values in Section 9 come from the simulated data and are demonstrations only.

3.15 Data Collection Procedure

Proposed procedure: eligible students receive an online link; consent is obtained before any question; the six scenarios are presented in randomised order; Stage B cards follow after all

Stage A choices are completed; respondents receive an anonymous identifier. Typical completion time is expected to be 12–15 minutes (to be verified in the pilot).

3.16 Data Cleaning

Responses would be screened for completeness, failed attention checks, speeding and straight-lining. Missing data below 5% per scale would be handled by listwise deletion for composite scores. Distributions would be inspected for outliers and data-entry errors. No cleaning has been performed on genuine data because none exists.

3.17 Statistical Analysis Plan

All tests are two-tailed with $\alpha = .05$. Descriptive statistics (frequency, percentage, mean, median, SD, minimum, maximum) address RQ1, RQ2 and RQ5. Cronbach's alpha assesses reliability. Shapiro–Wilk tests and skewness/kurtosis guide the choice between parametric and non-parametric procedures. Because Likert-derived scores are ordinal-like and frequently non-normal, Spearman correlation is the default for association. Group comparisons use Welch's t-test when both groups are approximately normal and the Mann–Whitney U test otherwise; a Kruskal–Wallis test compares intention across current-mode groups. A binary logistic regression addresses RQ3, and a conditional (multinomial-logit-type) model examines the role of option attributes in Stage A choices. Baseline and post-recommendation choices are compared with the Wilcoxon signed-rank test on respondent-level shares and McNemar's exact test on paired trip-level decisions (acknowledging clustering within respondents); modelled CO_2 differences use a paired t-test or Wilcoxon test depending on normality of the differences. Family-wise error across the seven hypothesis tests is described with Holm-adjusted p-values. Effect sizes accompany tests.

Table 3: Research question–variable–analysis mapping.

RQ	Variables	Analysis	Decision rule
RQ1	`awareness_score`; `concern_score`	Descriptives; Wilcoxon vs midpoint 3	$p < .05$
RQ2	`intention_score`; `private_vehicle_user`	Descriptives; Welch t / Mann–Whitney U	$p < .05$
RQ3	`low_emission_majority`; six predictors	Binary logistic regression	LR test $p < .05$; Wald tests
RQ4	awareness, concern, intention, `shift_count`	Spearman ρ	$p < .05$
RQ5	`acceptance_score`; `convenience_score`	Descriptives; Spearman ρ	$p < .05$
RQ6	`car_share`, `estimated_co2_saved`	Wilcoxon; McNemar; paired t / Wilcoxon	$p < .05$

3.18 Ethical Considerations

Participation would be voluntary, with informed consent obtained at the start and the right to withdraw. Responses are anonymous and used only for academic purposes; no passwords, financial details or directly identifying information are collected. Data would be stored securely. Institutional or supervisor approval will be obtained where required before any genuine data collection. No human participants were involved in the present paper, because all numerical data are simulated.

4 Research Instrument (Questionnaire)

This section summarises the proposed questionnaire; the full participant-facing text, with response anchors, is in Appendix A and the scenario cards in Appendix B. The instrument is proposed, has not been piloted and has not been administered to any participant.

4.1 Section A — Demographic and Commuting Profile

Age (years); gender; year of study; programme; one-way commute-distance category (< 2 km, 2–5 km, 5–10 km, > 10 km); main current commuting mode (private car, two-wheeler, bus [public or university], walking/cycling, carpool/shared ride); and commuting days per week.

4.2 Section B — Sustainable Mobility Awareness

Nine agreement items (1 = strongly disagree to 5 = strongly agree) plus one unscored attention-check item. Items address knowledge of low-emission options, awareness of the contribution of private cars, active and shared modes, the grid-dependence of EV benefits, digital tools and the ability to compare CO_2 impact (items aw1–aw9, Appendix A).

4.3 Section C — Environmental Concern and Attitudes

Seven agreement items (ec1–ec7) covering concern about air pollution and climate change, personal responsibility, belief in the effect of individual choices, institutional encouragement and willingness to accept a slightly longer trip for lower emissions.

4.4 Section D — Commuting Decision Factors

Twelve agreement items: three each for cost sensitivity (cost1–3), time sensitivity (time1–3) and perceived convenience (conv1–3), and one single item each for comfort, safety and accessibility.

4.5 Section E — Sustainable Travel Intention

Seven willingness items (1 = very unwilling to 5 = very willing) for the next semester: walking, cycling, public transport, carpooling, shared mobility, e-bike/e-scooter and electric vehicles.

4.6 Section F — Scenario-Based Commute Choice (Stage A)

Six scenarios (S1–S6) as summarised in Table 4. In each, the respondent selects the option they would choose for the trip described.

Table 4: Overview of scenario-based commute-choice tasks (design values; full option cards in Appendix B).

ID	Context	Distance	Options presented
S1	Short campus-area trip, fair weather, off-peak	3 km	Private car, City bus, Bicycle, E-bike, Shared ride
S2	Peak-hour trip, congested corridor	6 km	Private car, City bus, Bicycle, Carpool, E-bike
S3	Hot, humid day, medium distance	8 km	Private car, City bus, Metro/rail, Carpool, Shared ride
S4	Rainy day, longer distance	12 km	Private car, City bus, Metro/rail, Carpool, Shared ride
S5	Long trip with first/last-mile leg (multimodal)	15 km	Private car, Metro + e-bike, City bus, Carpool, Shared ride
S6	Evening return trip, personal-safety concern	5 km	Private car, City bus, Carpool, Shared ride, E-bike

4.7 Section G — Smart Commute Assistant Acceptance

Part G1 shows, for each scenario, a personalised recommendation card (Stage B) and records the post-recommendation choice (Stage C). Part G2 contains seven agreement items on perceived usefulness, trust, recommendation acceptance, willingness to follow suggestions, perceived environmental value, intention to continue use and intention to check carbon feedback. The first six form the acceptance scale (the item on continued use is shared with the user-engagement index).

5 Operational Definitions of Variables

Table 5 maps each construct to its variable, measurement, scale, basis in the literature and the analysis in which it is used. Composite scores are the arithmetic mean of their items and therefore range from 1 to 5.

Table 5: Constructs, variables, operational definitions and analysis.

Construct	Variable	Measurement	Scale	Basis	Analysis
Environmental awareness	`awareness_score`	Mean of aw1–aw9	1–5	Eslami (2025); Bamberg & Möser (2007)	RQ1, RQ3, RQ4; t-test
Environmental concern	`concern_score`	Mean of ec1–ec7	1–5	Bamberg & Möser (2007)	RQ1, RQ3, RQ4; Spearman
Sustainable travel intention	`intention_score`	Mean of int1–int7	1–5	Bamberg & Möser (2007); Lättman & Otsuka (2024)	RQ2, RQ4; t-test, Kruskal–Wallis
Cost sensitivity	`cost_sensitivity`	Mean of cost1–3	1–5	Wu et al. (2025)	RQ3; logistic regression
Time sensitivity	`time_sensitivity`	Mean of time1–3	1–5	Wu et al. (2025)	RQ3; logistic regression
Perceived convenience	`convenience_score`	Mean of conv1–3	1–5	Smith et al. (2023)	RQ3, RQ5; Spearman
Comfort, safety, accessibility	`comfort`, `safety`, `access`	Single items	1–5	Cui & Zhang (2024); Marzolla et al. (2025)	Descriptives
Recommendation acceptance	`acceptance_score`	Mean of acc1–acc6	1–5	Eslami (2025); Niazi et al. (2024)	RQ5; Spearman
User engagement	`engagement`	Mean of acc6, acc7	1–5	Niazi et al. (2024); Sunio (2018)	Descriptives
Sustainable mode choice	`low_emission_majority`	1 if ≥ 4 of 6 Stage A choices are non-car	0/1	Comi & Polimeni (2024)	RQ3; logistic regression
Mode shift	`shift_count`, `mode_shift_indicator`	Number of car→non-car changes	Count / 0/1	Eslami (2025)	RQ4; Spearman
Estimated CO_2 savings	`estimated_co2_saved`	Baseline − recommended emission (g)	Continuous	Ndhlovu et al. (2025)	RQ6; paired t/Wilcoxon

6 Data Dictionary

Table 6 lists the respondent-level variables. Likert items are integers from 1 to 5 with no reverse-keyed items. Composite scores are arithmetic means. Scenario variables store a single-letter mode code: C private car, B city bus, M metro/rail, Y bicycle, E e-bike, P carpool, S shared ride, X metro+e-bike (Appendix C).

Table 6: Respondent-level data dictionary (simulated dataset structure).

Variable	Type / range	Description
`respondent_id`	String	Anonymous identifier (R001–R120)
`age`	Integer 18–23	Age in years
`gender`	Categorical	Male / Female / Other-prefer not to say
`year_of_study`	Integer 1–4	Current year of undergraduate study

Variable	Type / range	Description
programme	Categorical	CSE / Other engineering / Management-Commerce / Sciences-Other
commute_distance_cat	Categorical	<2 / 2–5 / 5–10 / >10 km (one way)
current_mode	Categorical	Private car / Two-wheeler / Bus / Walking-Cycling / Carpool-Shared
commute_days	Integer 3–6	Days per week travelling to campus
aw1–aw9	Likert 1–5	Sustainable-mobility awareness items
ec1–ec7	Likert 1–5	Environmental-concern items
cost1–cost3	Likert 1–5	Cost-sensitivity items
time1–time3	Likert 1–5	Time-sensitivity items
conv1–conv3	Likert 1–5	Convenience items
comfort, safety, access	Likert 1–5	Single-item comfort, safety and accessibility
int1–int7	Likert 1–5	Sustainable travel intention items (walk, cycle, PT, car-pool, shared, e-bike, EV)
acc1–acc7	Likert 1–5	Assistant acceptance and engagement items
attention_check	Integer 1–5	Unscored quality-control item (correct = 4)
stageA_S1–stageA_S6	Mode code	Baseline (pre-recommendation) choice per scenario
rec_S1–rec_S6	Mode code	Personalised recommended mode per scenario
stageC_S1–stageC_S6	Mode code	Post-recommendation choice per scenario
emission_A_S1–emission_A_S6	Numeric (g)	Modelled CO_2 for the baseline choice
emission_C_S1–emission_C_S6	Numeric (g)	Modelled CO_2 for the post-recommendation choice
awareness_score	Numeric 1–5	Mean of aw1–aw9
concern_score	Numeric 1–5	Mean of ec1–ec7
cost_sensitivity	Numeric 1–5	Mean of cost1–cost3
time_sensitivity	Numeric 1–5	Mean of time1–time3
convenience_score	Numeric 1–5	Mean of conv1–conv3
intention_score	Numeric 1–5	Mean of int1–int7
acceptance_score	Numeric 1–5	Mean of acc1–acc6
engagement	Numeric 1–5	Mean of acc6 and acc7
total_awareness_score	Numeric 9–45	Sum of aw1–aw9 (for reporting)
total_intention_score	Numeric 7–35	Sum of int1–int7
total_acceptance_score	Numeric 6–30	Sum of acc1–acc6
private_vehicle_user	0/1	1 if current mode is private car or two-wheeler
low_emission_majority	0/1	1 if ≥ 4 of 6 Stage A choices are non-car
car_share_A, car_share_C	Numeric 0–1	Proportion of the six choices that are car
sust_share_A, sust_share_C	Numeric 0–1	Proportion of the six choices that are non-car
baseline_emission	Numeric (g)	Mean modelled CO_2 across the six baseline choices
recommended_emission	Numeric (g)	Mean modelled CO_2 across the six post-recommendation choices
estimated_co2_saved	Numeric (g)	baseline_emission − recommended_emission
shift_count	Integer 0–6	Number of scenarios changing from car to non-car
mode_shift_indicator	0/1	1 if shift_count ≥ 1
selected_mode	Mode code	Representative selected mode (Stage C, S1)

7 Emission Model

Emissions are estimated with a transparent distance–factor model. For a trip made by a single mode m,

$$E_m = d \times EF_m, \tag{1}$$

where E_m is the estimated emissions for mode m (grams of CO_2), d is the passenger travel distance (kilometres) and EF_m is the emission factor for mode m (grams of CO_2 per passenger-kilometre). For a route comprising several legs,

$$E_{\text{total}} = \sum_i (d_i \times EF_i)\,. \tag{2}$$

Relative to a private-car baseline, the modelled saving is

$$CO_2 \text{ saved} = E_{\text{car}} - E_{\text{selected}}. \tag{3}$$

The multimodal metro+e-bike option (code X) is evaluated with Equation 2 as a distance-weighted combination of its legs.

The emission factors used in the simulated demonstration are listed in Table 7. They are **illustrative model inputs only**. Real implementation would require region-specific, validated emission factors that reflect local fleet composition, occupancy, fuel type and grid mix; the electric-mode factor in particular depends on how electricity is generated (Ndhlovu et al., 2025). Bus, carpool and shared-ride factors represent emissions allocated per passenger under assumed occupancies. No universal fixed emission value is claimed.

Table 7: Illustrative emission factors used in the simulated demonstration (g CO_2 per passenger-km). These are model inputs, not measured values for the Chandigarh region.

Code	Mode	EF_m (g/pkm)	Note
C	Private car (solo)	170	Reference baseline
B	City bus	70	Allocated per passenger, assumed occupancy
M	Metro / rail	30	Allocated per passenger
Y	Bicycle	0	No tailpipe emissions modelled
E	E-bike / e-scooter	20	Grid-dependent in reality
P	Carpool	57	Car factor ÷ assumed occupancy 3
S	Shared ride	60	Car factor ÷ assumed occupancy ≈ 2.8
X	Metro + e-bike	42	Distance-weighted multimodal combination

Applying Equation 1 to each scenario's distance gives the all-car reference emissions shown in Table 21 (Section 9); for example, the 8 km scenario (S3) has an all-car reference of $8 \times 170 = 1360$ g.

8 Simulated Demonstration Dataset

Purpose and status. A synthetic dataset of $N = 120$ respondents was generated with a computer program (Python; NumPy; fixed random seed `20251006`) to demonstrate the analysis procedure. It is *not* real data. The respondents are fictional; they are not Chandigarh University students and no real person provided any value. The generation script and the resulting dataset accompany this paper so that the demonstration is fully reproducible.

Generation procedure. (1) Demographics and commute-distance categories were drawn from assumed distributions, and the current mode was drawn conditional on distance. (2) Latent traits (awareness, concern, convenience, cost sensitivity, time sensitivity, intention, acceptance and engagement) were drawn from correlated normal distributions with modest offsets for private-vehicle users. (3) Likert items were generated from latent traits plus independent noise, rounded and clipped to 1–5, so that scales are imperfectly reliable and values are not perfectly correlated. (4) Stage A scenario choices were generated from a random-utility (multinomial-logit-type) model with terms for travel time, cost, comfort, convenience, emissions and a current-mode inertia term. (5) Personalised recommendations were generated by ranking options with respondent-specific weights on time, cost and CO_2. (6) Stage C choices followed the recommendation with a probability that increased with the acceptance score and, weakly, with environmental concern; otherwise the Stage A choice was retained.

Consequences for interpretation. Associations such as awareness–intention and convenience–acceptance were deliberately built into the generator, so detecting them demonstrates that the procedure works and is not evidence about real commuters. The weak concern effect on following recommendations was built in at a small size, so a

non-significant result illustrates limited power for small effects. Where the generator contains no effect, test results show how null results would be reported. The recommendation rule never recommends an option worse than all alternatives on its own weighted criteria, and the generator allows Stage C to differ from Stage A only through following a recommendation, so patterns such as a decline in car share are partly built in. Agreement with any published study would likewise be by construction and carries no evidential weight.

Plausibility checks. Scale means lie between 3.46 and 3.65 (not at ceiling), scale reliabilities range from the questionable to the good range, correlations are moderate rather than near-perfect, and baseline private-car share (35.0% of decisions) is plausible for a scenario set that includes short and long trips. These checks establish internal plausibility only. The complete respondent-level dataset is provided as an accompanying data file (Appendix E); a representative extract is shown in Table 28.

9 Results and Data Analysis

The numerical results in this section are generated from a simulated demonstration dataset and are intended solely to illustrate the proposed analysis procedure. They do not represent actual students or actual system deployment outcomes.

All analyses use the $N = 120$ simulated respondents of Appendix E and were computed in Python (pandas, SciPy, statsmodels). Tests are two-tailed with $\alpha = .05$. Because the data were generated with built-in structure (Section 8), significance here demonstrates that the analytical procedure functions and does not support any claim about real commuters or any real system.

9.1 Simulated Participant Profile

The simulated respondents were aged 18–23 years ($M = 19.86$, $SD = 1.29$). 58.3% were male, 38.3% female and 3.3% other/prefer not to say. 42.5% mainly used a private car or two-wheeler. Table 8 and Figure 2 show the profile.

Table 8: Simulated participant profile (simulated demonstration data, $N = 120$).

Variable	Category	n	%
Gender	Male	70	58.3%
	Female	46	38.3%
	Other/Prefer not to say	4	3.3%
Year of study	1	34	28.3%
	2	45	37.5%
	3	31	25.8%
	4	10	8.3%
Programme	CSE	47	39.2%
	Other engineering	38	31.7%
	Management/Commerce	27	22.5%
	Sciences/Other	8	6.7%
Commute distance (one-way)	<2 km	26	21.7%
	2–5 km	33	27.5%
	5–10 km	33	27.5%
	>10 km	28	23.3%
Current mode	Private car	22	18.3%
	Two-wheeler	29	24.2%
	Bus (public/university)	24	20.0%
	Walking/Cycling	26	21.7%
	Carpool/Shared ride	19	15.8%
Commuting days per week	3	10	8.3%
	4	17	14.2%
	5	49	40.8%
	6	44	36.7%

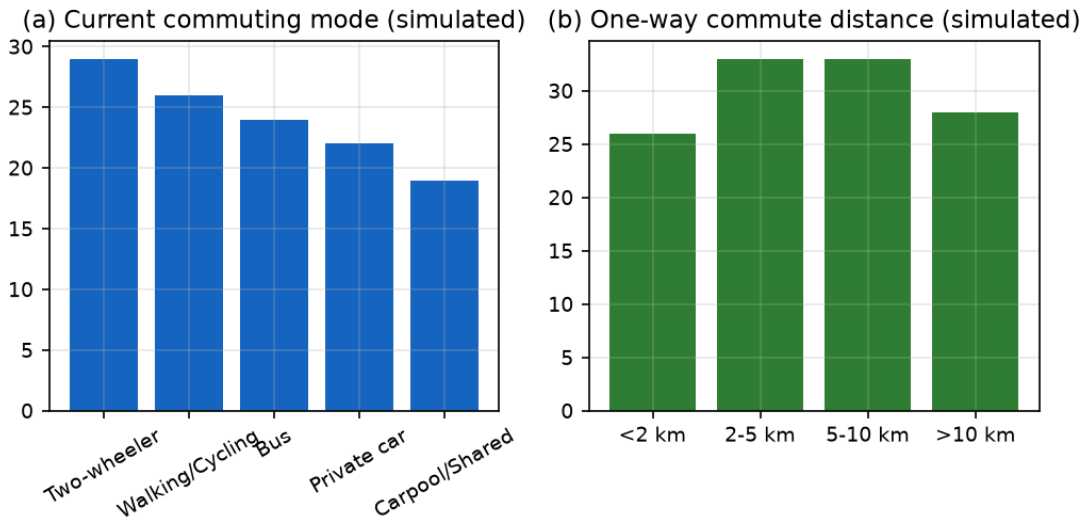


Figure 2: Respondent and commuting profile of the simulated demonstration dataset ($N = 120$): (a) current commuting mode; (b) one-way commute distance.

9.2 Sustainable Mobility Awareness

Mean awareness was 3.52 ($SD = 0.76$; median = 3.56; range 1.67–4.89), above the scale midpoint of 3 (one-sample Wilcoxon signed-rank, $p < .001$). 30.0% of simulated respondents scored 4.00 or higher. Table 9 gives item-level results and Figure 3 the distribution. RQ1 is therefore illustrated as moderate-to-high simulated awareness; this reflects generator settings and is not an estimate for real students.

Table 9: Awareness items (simulated demonstration data, $N = 120$).

Item	*M*	*SD*	Agree (4–5)
Knows which options emit less CO_2 than a car	3.51	1.14	51.7%
Understands car commuting contributes to emissions	3.61	1.25	54.2%
Aware public transport can emit less per passenger-km	3.49	1.05	50.0%
Knows walking/cycling have almost no tailpipe emissions	3.48	1.19	51.7%
Aware carpooling reduces vehicles on the road	3.52	1.10	50.0%
Aware shared e-bikes/e-scooters can replace short car trips	3.55	1.05	52.5%
Understands EV benefit depends on electricity source	3.51	1.04	52.5%
Knows apps showing multimodal routes/emissions	3.58	1.15	56.7%
Can judge which option has lower CO_2	3.43	1.14	57.5%

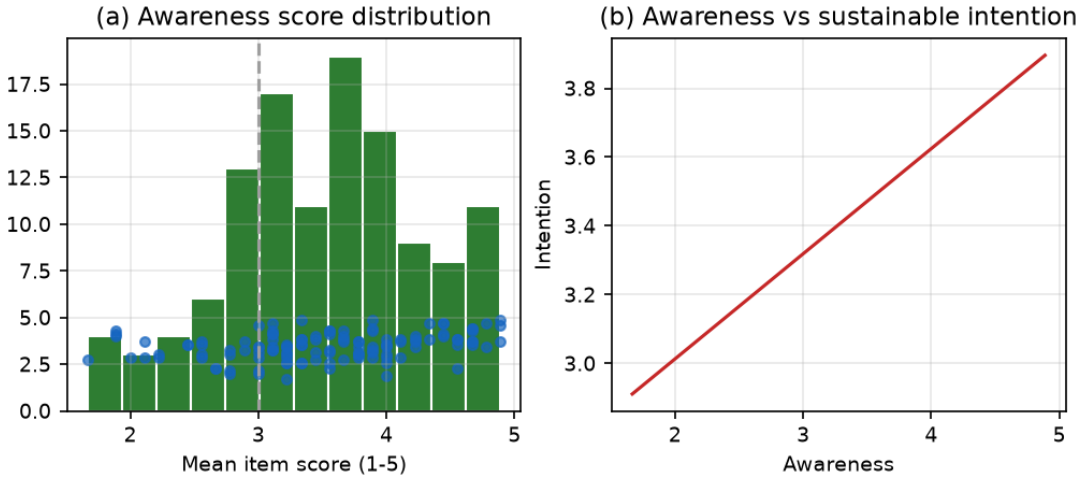


Figure 3: (a) Distribution of environmental awareness scores in the simulated dataset ($N = 120$); (b) awareness versus sustainable travel intention, with a linear trend line for illustration.

9.3 Environmental Concern and Attitudes

Mean environmental concern was 3.65 ($SD = 0.72$; median = 3.71). Agreement was highest for "feels personal responsibility for travel emissions" and lowest for "concerned about transport emissions and climate" (Table 10).

Table 10: Environmental-concern items (simulated demonstration data, $N = 120$).

Item	M	SD	Agree (4–5)
Concerned about road-traffic air pollution	3.64	1.09	60.0%
Concerned about transport emissions and climate	3.52	1.20	55.0%
Feels personal responsibility for travel emissions	3.76	1.08	60.8%
Uncomfortable choosing high-emission option when cleaner exists	3.69	1.04	54.2%
Believes individual choices make a difference	3.65	1.03	55.8%
Universities should encourage low-emission commuting	3.66	1.03	55.8%
Emission reduction important even if slightly slower	3.62	1.05	57.5%

9.4 Commuting Behaviour and Decision Factors

In the simulated data, convenience ($M = 3.55$, $SD = 0.76$), time sensitivity ($M = 3.53$, $SD = 0.80$) and cost sensitivity ($M = 3.46$, $SD = 0.86$) were the stated decision factors, with single-item comfort, safety and accessibility ratings shown in Table 11. Current-mode shares are in Figure 2. Full descriptive statistics for all composite scores, including median, range, skewness, excess kurtosis and the Shapiro–Wilk test, are given in Table 12.

Table 11: Commuting decision factors (simulated demonstration data, $N = 120$).

Factor	M	SD	Median
Cost sensitivity (3 items)	3.46	0.86	3.67
Time sensitivity (3 items)	3.53	0.80	3.50
Perceived convenience (3 items)	3.55	0.76	3.67
Comfort (1 item)	3.70	1.00	4.00
Safety (1 item)	3.62	1.05	4.00
Accessibility (1 item)	3.61	1.04	4.00

Table 12: Descriptive statistics and distribution assessment of composite scores (simulated demonstration data, $N = 120$).

Score	M	SD	Mdn	Min	Max	Skew	Kurt.	SW p
Awareness (aw)	3.52	0.76	3.56	1.67	4.89	−0.22	−0.43	.078
Concern (ec)	3.65	0.72	3.71	1.57	5.00	−0.41	−0.26	.034
Cost sensitivity	3.46	0.86	3.67	1.00	5.00	−0.29	−0.32	.013
Time sensitivity	3.53	0.80	3.50	1.33	5.00	−0.33	−0.46	.002
Convenience	3.55	0.76	3.67	1.33	5.00	−0.12	−0.50	.003
Intention	3.48	0.72	3.57	1.71	4.86	−0.18	−0.41	.184
Acceptance	3.56	0.72	3.67	1.83	5.00	−0.33	−0.42	.039
Engagement	3.54	0.88	3.50	1.50	5.00	−0.32	−0.44	$< .001$

9.5 Sustainable Travel Intention

Mean sustainable travel intention was 3.48 ($SD = 0.72$; median = 3.57), above the midpoint (Wilcoxon $p < .001$). Willingness was highest for walking and lowest for carpooling (Table 13).

Table 13: Mode-specific sustainable travel intention (simulated demonstration data, $N = 120$).

Mode	M	SD	Willing (4–5)
Walking	3.57	1.06	51.7%
Cycling	3.46	1.11	55.0%
Public transport	3.50	1.13	53.3%
Carpooling	3.41	1.05	47.5%
Shared mobility	3.48	1.07	55.0%
E-bike/e-scooter	3.42	1.07	50.0%
Electric vehicle	3.51	1.17	58.3%

9.6 Reliability Analysis

Cronbach's alpha was computed on the simulated items for each multi-item scale (Table 14). Awareness ($\alpha = .85$), concern ($\alpha = .79$), intention ($\alpha = .78$) and acceptance ($\alpha = .74$) reached the conventional benchmark of .70; the three-item cost ($\alpha = .68$), time ($\alpha = .58$) and convenience ($\alpha = .54$) scales fell below it. These values are demonstrations calculated from simulated data; they do not establish the reliability of the instrument with real respondents, and the short three-item scales would require review before the main survey.

Table 14: Reliability of multi-item scales (simulated demonstration data, $N = 120$).

Scale	Items	Cronbach's α
Awareness	9	.85
Environmental concern	7	.79
Sustainable travel intention	7	.78
Assistant acceptance	6	.74
Cost sensitivity	3	.68
Time sensitivity	3	.58
Convenience	3	.54

9.7 Correlation Analysis

Awareness and intention were positively correlated (Spearman $\rho = .34$, $p < .001$; H1). Convenience and assistant acceptance were weakly-to-moderately positively correlated ($\rho = .21$, $p = .023$; H3). Environmental concern and the number of car-to-sustainable shifts after recommendation were not significantly correlated ($\rho = .03$, $p = .785$; H2). Correlation does not imply causation; relationships here were partly built into the generator. Table 15 shows the full matrix and Figure 3(b) the awareness–intention relationship.

Table 15: Spearman correlation matrix (simulated demonstration data, $N = 120$). $^{*}p < .05$, $^{**}p < .01$, $^{***}p < .001$.

	Aware.	Concern	Intent.	Conv.	Cost	Time	Accept.
Awareness	—	.23*	.34***	−.15	−.14	.20*	.14
Concern	.23*	—	.22*	−.20*	−.14	−.10	.16
Intention	.34***	.22*	—	−.15	−.10	.03	.12
Convenience	−.15	−.20*	−.15	—	−.05	−.02	.21*
Cost sens.	−.14	−.14	−.10	−.05	—	.03	−.16
Time sens.	.20*	−.10	.03	−.02	.03	—	.10
Acceptance	.14	.16	.12	.21*	−.16	.10	—

9.8 Group Comparisons

Three exploratory grouping comparisons were run. The private-vehicle comparison corresponds to H4. For each outcome, the test was chosen after Shapiro–Wilk checks of both groups (Table 16). Private-vehicle users scored 3.27 versus 3.63 for others on sustainable intention ($t(103.9) = -2.83$, $p = .006$, $d = -0.53$), so H4 was supported in this simulated sample. High-awareness respondents reported higher intention: the median split gave 3.66 versus 3.28 ($t(116.9) = 3.00$, $p = .003$, $d = 0.55$), which is expected from the correlation in Section 9 and from how the data were generated. Distance-group differences were small and non-significant. A Kruskal–Wallis test across the five current-mode groups was not significant ($H(4) = 8.36$, $p = .079$).

Table 16: Group comparisons (simulated demonstration data, $N = 120$). Cells show M (SD); share variables are on a 0–1 scale and CO_2 saved is g per trip.

Comparison	Outcome	Group 1	Group 2	Test	Effect
Private vehicle ($n = 51$) vs other ($n = 69$)	Sustainable intention	3.27 (0.72)	3.63 (0.68)	$t(103.9) = -2.83$, $p = .006$	$d = -0.53$
	Environmental awareness	3.46 (0.71)	3.56 (0.79)	$t(113.7) = -0.73$, $p = .468$	$d = -0.13$
	Assistant acceptance	3.54 (0.73)	3.57 (0.71)	$t(106.4) = -0.25$, $p = .803$	$d = -0.05$
Short <5 km ($n = 59$) vs long ≥5 km ($n = 61$)	Sustainable intention	3.51 (0.76)	3.45 (0.68)	$t(115.5) = 0.46$, $p = .643$	$d = 0.09$
	Car share at baseline	0.34 (0.27)	0.36 (0.23)	$U = 1666.0$, $p = .476$	$r_{rb} = 0.07$
	CO_2 saved/trip (g)	125.7 (168.9)	142.3 (191.3)	$U = 1684.5$, $p = .538$	$r_{rb} = 0.06$
High ($n = 62$) vs low awareness ($n = 58$)	Sustainable intention	3.66 (0.68)	3.28 (0.71)	$t(116.9) = 3.00$, $p = .003$	$d = 0.55$
	Assistant acceptance	3.70 (0.69)	3.41 (0.71)	$t(116.9) = 2.22$, $p = .028$	$d = 0.41$
	Baseline sustainable share	0.70 (0.23)	0.60 (0.27)	$U = 2159.5$, $p = .053$	$r_{rb} = 0.20$

9.9 Regression Analysis

A binary logistic regression predicted whether a respondent chose a non-car option in at least four of six baseline scenarios (outcome = 1 for 77 and 0 for 43 respondents). The six predictors were entered together (Table 17). The model was significant overall (likelihood-ratio $\chi^2(6) = 35.52$, $p < .001$; McFadden's pseudo-$R^2 = .23$; AUC = .83; correct classification at 0.5 = 76.7%). Individually significant predictors were perceived convenience (negative) and cost sensitivity (positive). Odds ratios are per one-point increase on the 1–5 scale; for example, each additional point of cost sensitivity multiplied the odds of a low-emission majority by 2.00 holding other predictors constant, whereas each additional point of perceived convenience multiplied the odds by 0.30. The smaller outcome class (43) gives about 7.2 events per predictor, below the common guideline of 10, so coefficients should be read cautiously. Coefficients describe associations in simulated data and not causal effects.

Table 17: Logistic regression predicting low-emission majority at baseline (simulated demonstration data, $N = 120$).

Predictor	B	SE	p	**OR**	**95% CI (OR)**
Intercept	2.743	2.594	.290	15.53	0.10–2508.01
Environmental awareness	0.508	0.338	.133	1.66	0.86–3.23
Environmental concern	0.496	0.334	.138	1.64	0.85–3.16
Perceived convenience	−1.210	0.344	< .001	0.30	0.15–0.59
Cost sensitivity	0.694	0.279	.013	2.00	1.16–3.46
Time sensitivity	−0.455	0.295	.122	0.63	0.36–1.13
Assistant acceptance	−0.574	0.335	.087	0.56	0.29–1.09

Attribute-based mode choice. To examine which option attributes drive Stage A choices, a conditional (multinomial-logit-type) model was estimated on the 720 choices with standardised attributes (Table 18). Travel time ($B = -1.11$, $p < .001$), cost ($B = -0.34$, $p = .022$), convenience ($B = 0.30$, $p = .006$) and comfort ($B = 0.19$, $p = .043$) were significant; emissions were not ($B = 0.04$, $p = .714$), consistent with the design decision to withhold CO_2 from Stage A.

Table 18: Conditional logit of Stage A mode choice on option attributes (simulated demonstration data; 720 choices).

Attribute (standardised)	B	SE	p
Travel time	−1.107	0.176	< .001
Cost	−0.340	0.148	.022
Comfort	0.188	0.093	.043
Convenience	0.298	0.108	.006
Emissions (CO_2)	0.039	0.106	.714

9.10 Baseline versus Recommendation Analysis (Simulated Behavioural Demonstration)

This is a *simulated behavioural demonstration*, not an actual experiment. Each simulated respondent made a Stage A choice in six scenarios, saw a personalised recommendation (Stage B) and chose again (Stage C), giving 720 paired decisions.

The private-car share of decisions fell from 35.0% at baseline to 25.6% after the recommendation, and the sustainable-mode share rose from 65.0% to 74.4% (Table 19; Figure 4). At respondent level, mean car share fell from 0.35 to 0.26 (Wilcoxon signed-rank $W = 0.0$, $p < .001$). On the paired trip-level decisions, 68 moved from car to a non-car option and 0 moved in the opposite direction (McNemar exact $p < .001$); because decisions are clustered within respondents, the respondent-level test is the primary inference. The recommendation differed from the baseline choice in 502 of 720 decisions and was followed in 141, a recommendation acceptance rate of 28.1%. 48 respondents (40.0%) shifted away from the car in at least one scenario, and 27.0% of baseline car decisions (68 of 252) became non-car decisions. Mean differences per trip were −0.15 min in travel time (Wilcoxon $p = .263$) and −10.30 rupees in cost ($p < .001$), i.e. a negligible time penalty and a modelled cost saving.

Table 19: Baseline versus post-recommendation mode share (simulated demonstration data; 720 paired decisions).

Mode	Stage A n	Stage A %	Stage C n	Stage C %	Change (pp)
Private car	252	35.0%	184	25.6%	−9.4
City bus	40	5.6%	27	3.8%	−1.8
Metro/rail	81	11.2%	123	17.1%	+5.8
Metro + e-bike	36	5.0%	64	8.9%	+3.9
Bicycle	35	4.9%	53	7.4%	+2.5
E-bike	112	15.6%	147	20.4%	+4.9
Carpool	82	11.4%	64	8.9%	−2.5
Shared ride	82	11.4%	58	8.1%	−3.3
Sustainable (any non-car)	468	65.0%	536	74.4%	+9.4

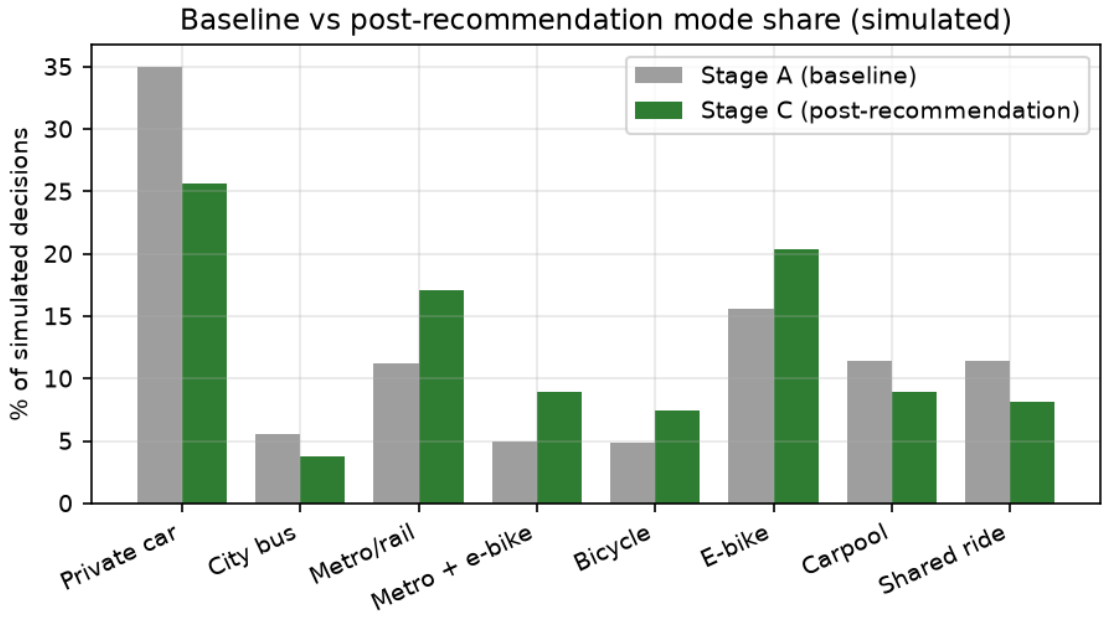


Figure 4: Baseline (Stage A) versus post-recommendation (Stage C) mode choice across 720 simulated decisions.

Table 20: Behavioural and trade-off indicators, baseline versus recommendation (simulated demonstration data).

Indicator	Value
Baseline private-car share	35.0%
Post-recommendation private-car share	25.6%
Baseline sustainable-mode share	65.0%
Post-recommendation sustainable-mode share	74.4%
Recommendation acceptance rate (141 of 502 differing recommendations)	28.1%
Mode-shift rate (baseline car decisions moved off car)	27.0% (68 of 252)
Respondents with at least one shift	48 (40.0%)
Mean travel-time difference per trip (Stage C − A)	−0.15 min
Mean cost difference per trip (Stage C − A)	−10.30 rupees
Mean CO_2 difference per trip (Stage C − A)	−134.1 g

Recommendations were personalised through respondent-specific weights on time, cost and CO_2, but the options in some scenarios were close enough that many respondents received the same recommendation; recommended modes were E-bike (215), Metro/rail (198), Bicycle (136), Metro + e-bike (117), Carpool (34), City bus (15) and Private car (5). The rule recommended the private car in only 5 decisions, so reverse shifts toward the car were possible but rare.

9.11 CO_2 Savings Analysis

Using the emission model (Section 7), baseline emissions averaged 717.0 g CO_2 per trip and post-recommendation emissions 582.8 g, a mean saving of 134.1 g per trip (95% CI 106.0–162.3 g; 18.7% reduction; paired $t(119) = -9.37$, $p < .001$; Cohen's $d_z = 0.35$). Because the paired differences were not normally distributed (Shapiro–Wilk $p < .001$), the Wilcoxon signed-rank test is preferred ($p < .001$) and leads to the same conclusion. Relative to an all-car reference, mean per-trip emissions would be 1388.3 g; baseline choices already avoid 671.3 g and post-recommendation choices 805.5 g of that reference. Table 21 and Figure 5 show results by scenario.

Table 21: Modelled CO_2 per trip by scenario (g; simulated demonstration data).

Scenario	Distance	All-car ref.	Stage A mean	Stage C mean	Saving	Reduction
S1	3 km	510	236.2	187.2	49.0	20.7%
S2	6 km	1020	443.9	334.2	109.7	24.7%
S3	8 km	1360	722.4	593.0	129.4	17.9%
S4	12 km	2040	1071.1	873.0	198.1	18.5%
S5	15 km	2550	1396.1	1150.9	245.2	17.6%
S6	5 km	850	432.2	358.8	73.4	17.0%
All	—	1388	717.0	582.8	134.1	18.7%

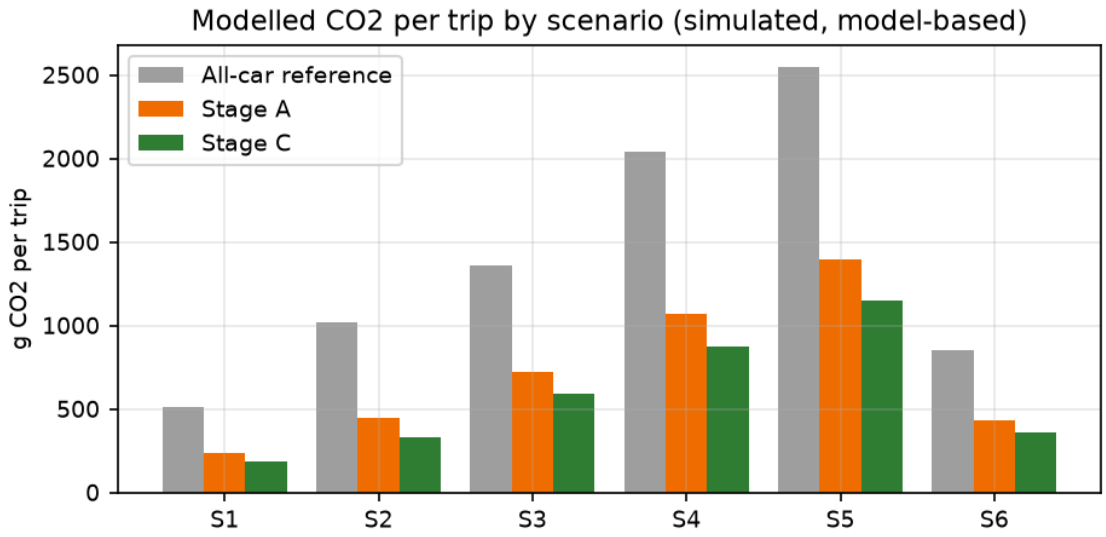


Figure 5: Modelled CO_2 per trip by scenario: all-car reference, simulated baseline (Stage A) and post-recommendation (Stage C). Model-based illustration only.

9.12 Illustrative Environmental Projection Based on Simulated Data

This projection is **model-based and illustrative**; it is not observed environmental impact. Assumptions: (i) the six scenarios are equally weighted representative trips with a mean distance of 8.2 km (the mean of the six scenario distances; they are not each respondent's own commute); (ii) the mean simulated saving per trip of 134.1 g CO_2 persists; (iii) a commuter makes 40 trips per month (20 commuting days × 2 trips); (iv) an academic year of 9 months (360 trips); (v) the emission factors of Table 7 hold; (vi) a hypothetical cohort of 1,000 commuters behaves like the simulated sample. Because stated-choice responses commonly overstate real behaviour change, sensitivity rows apply 50% and 25% of the simulated effect.

Table 22: Illustrative CO_2 projection from simulated data (model-based; not observed).

Scenario	g/trip	kg/month	kg/year	tonnes/1,000 commuters/year
Baseline emissions (reference)	717.0	28.7	258.1	258.1
100% of simulated effect	134.1	5.4	48.3	48.3
50% of simulated effect	67.1	2.7	24.1	24.1
25% of simulated effect	33.5	1.3	12.1	12.1

Under the full simulated effect, savings would be 5.4 kg CO_2 per commuter per month and 48.3 kg per academic year, or 48.3 tonnes for 1,000 commuters. These figures depend entirely on the simulation inputs and should not be quoted as expected real-world savings (Figure 6).

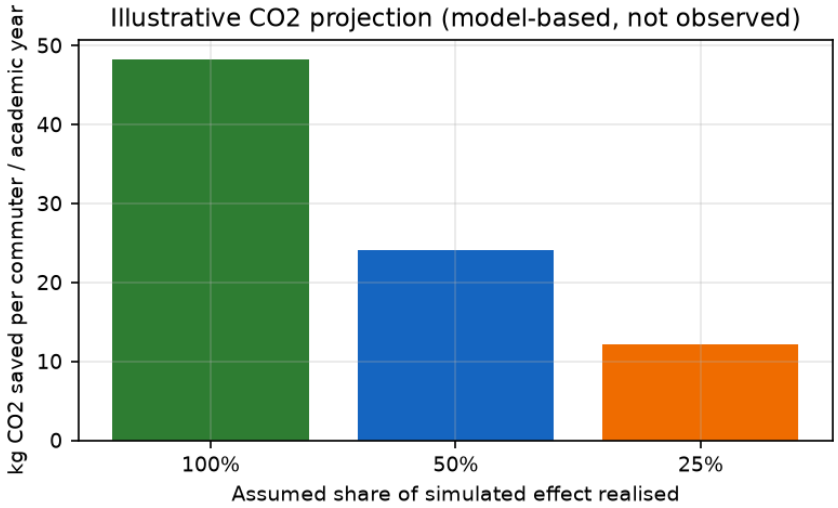


Figure 6: Illustrative CO_2 projection under assumed shares of the simulated effect (model-based, not observed).

9.13 Hypothesis Testing Summary

Six hypotheses (seven tests) are summarised in Table 23. Results are demonstrations on simulated data. Holm-adjusted p-values for the seven tests were: H1 = .001, H2 = .785, H3 = .045, H4 = .017, H5 < .001, H6a < .001, H6b < .001. H2 was not supported; all other hypotheses were supported in the simulated data.

Table 23: Hypothesis testing summary (simulated demonstration data; $N = 120$).

Hypothesis	Test	Simulated result	p	Holm p	Decision
H1: awareness → intention	Spearman ρ	$\rho = .34$	$< .001$	.001	Supported
H2: concern → shift count	Spearman ρ	$\rho = .03$	.785	.785	Not supported
H3: convenience → acceptance	Spearman ρ	$\rho = .21$	.023	.045	Supported
H4: private-vehicle users lower intention	Welch t	$t = -2.83$	.006	.017	Supported
H5: predictors jointly predict choice	Logistic LR	$\chi^2(6) = 35.52$	$< .001$	$< .001$	Supported
H6a: recommendation reduces car share	Wilcoxon	$W = 0.0$	$< .001$	$< .001$	Supported
H6b: recommendation reduces CO_2/trip	Paired t	$t = -9.37$	$< .001$	$< .001$	Supported

10 Conceptual AI Framework: Eco-Friendly Daily Commute Helper

Status. The framework is conceptual. It has not been implemented, deployed or empirically validated. It retains the five-layer architecture of the original paper and connects it to the research variables, the behavioural task and the evaluation metrics of this study. The conceptual chain is: literature → research gap → research questions → variables → instrument → statistical analysis → simulated findings → behavioural and emission evaluation → AI framework → future validation (Figure 1).

The proposed helper has five layers (Figure 7), each tied to variables measured in the proposed study.

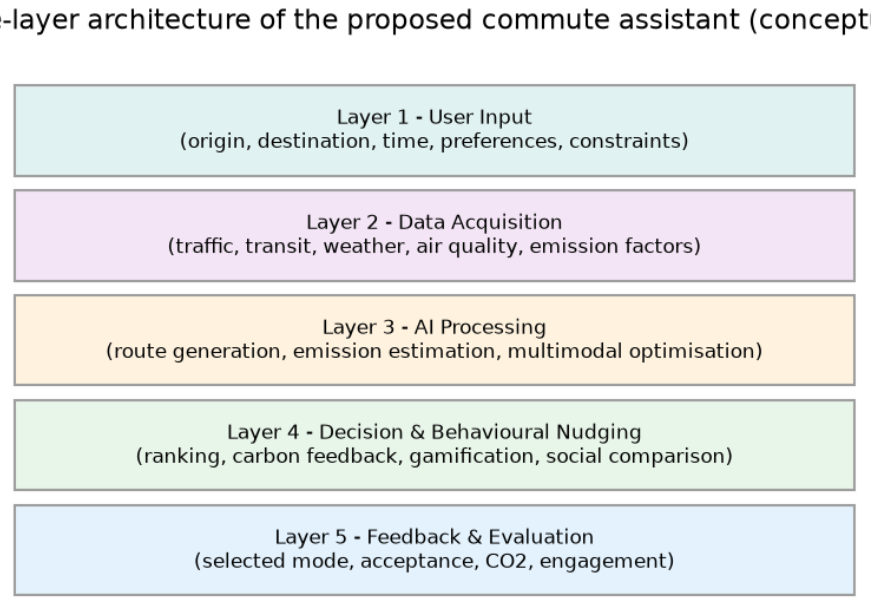


Figure 7: Five-layer architecture of the proposed AI-driven, behaviourally aware Eco-Friendly Daily Commute Helper (conceptual).

10.1 Layer 1 — User Input

Collects origin, destination, departure time and user preferences (cost, time, comfort, accessibility). In the research design, these correspond to `commute_distance`, `commute_days`, `current_mode` and the decision-factor scores.

10.2 Layer 2 — Data Acquisition

Gathers real-time traffic, public-transit schedules, weather and air-quality data, and emission factors for each mode (Kashem et al., 2025; Ouyang et al., 2025). Emission factors would be regional and updated as grid and fleet data change (Ndhlovu et al., 2025). In the simulation, scenario times, costs and emission factors are fixed design values.

10.3 Layer 3 — AI Processing

Generates multimodal route options (walking, cycling, public transport, shared mobility, electric vehicles), estimates emissions with Equations 1–3, and predicts congestion and energy use. This layer corresponds to the scenario option cards and the emission model of Section 7.

10.4 Layer 4 — Decision and Behavioural Nudging

Ranks options on a multi-criteria basis (CO_2, travel time, cost, health benefit) with weights that can be personalised, and reinforces low-emission choices through carbon feedback, gamified rewards and social comparison. In the simulated Stage B, the ranking used respondent-specific weights on time, cost and CO_2 and displayed a recommendation card; gamification and social comparison were not simulated.

10.5 Layer 5 — Feedback and Evaluation

Logs the selected mode, measures impact (CO_2 saved, mode shift, engagement) and feeds results back to personalise later recommendations. In the simulation, Stage C choices and the derived indicators play this role; no learning loop was simulated.

10.6 Mapping Research Variables to System Components

Table 24 links research variables to system inputs, AI processes, behavioural interventions and evaluation metrics. It describes design intent; it does not show that the framework works.

Table 24: Mapping of research variables to system inputs, AI processes, behavioural interventions and evaluation metrics.

Research variable	**System input (layer)**	**AI process (L3)**	**Behavioural intervention (L4)**	**Evaluation metric**
`commute_distance`; `current_mode`; `commute_days`	Origin, destination, time, usual mode (L1)	Multimodal route generation; distance-based option filtering	Mode-specific suggestions (e.g., bicycle on short trips)	Mode-shift rate; sustainable-mode share
`cost_sensitivity`; `time_sensitivity`; `convenience_score`; comfort; safety; accessibility	Preference weights (L1)	Multi-criteria ranking with personalised weights	Rank and CO_2 display balancing time, cost, comfort	Cost and time difference; acceptance
`awareness_score`	Optional profile baseline (L1)	Depth of explanation between options	CO_2 comparison cards	Awareness change (pre/post)
`concern_score`	Optional profile baseline (L1)	Message framing around responsibility	Carbon feedback framed as environmental benefit	Mode-shift rate; concern change
`intention_score`	Stated willingness by mode (L1)	Candidate-option filtering	Targeted prompts for intention change	Intention change; mode-shift rate
Traffic, transit, weather, air quality, emission factors	Real-time feeds and EF (L2)	Emission estimation ($E = d \times EF$); congestion prediction	Per-option CO_2 display	`baseline_emission`; `recommended_emission`; CO_2 saved
`acceptance_score` (usefulness, trust, acceptance)	Interface feedback (L5)	Explanation of ranking	Transparency of recommendation card	Acceptance; continued use; trust
`engagement`	Usage logs (L5)	Personalised choice prompts	Gamified rewards; social comparison	User engagement; retention

11 Evaluation Metrics

The original paper proposed five metrics. They are retained and extended with four metrics that capture attitudes and trust. Table 25 states what can be measured with the proposed survey and what requires a real deployment or longitudinal follow-up. In this paper, only the metrics marked "simulated" were computed, and only on simulated data.

Table 25: Evaluation metrics and measurability.

Metric	Definition	Measurable in proposed survey?	Status in this paper
Mode-shift rate	Share of baseline car decisions changed to a non-car option	Partly (stated choice)	Simulated
Sustainable-mode share	Share of decisions that are non-car	Yes	Simulated
Recommendation acceptance	Followed recommendations ÷ differing recommendations	Partly (stated choice)	Simulated
CO_2 saved per trip	$E_{\text{car}} - E_{\text{selected}}$ (Section 7)	Model-based only	Simulated
User engagement	Self-reported intention to continue use and check feedback	Yes (self-report)	Simulated
Awareness change	Pre/post change in `awareness_score`	Requires pre/post design	Not computed
Concern change	Pre/post change in `concern_score`	Requires pre/post design	Not computed
Trust in recommendations	Self-reported trust item (acc2)	Yes (self-report)	Simulated
Intention change	Pre/post change in `intention_score`	Requires pre/post design	Not computed

12 Discussion

Four kinds of statement are kept apart below: published evidence (cited), simulated results (from the demonstration dataset), the proposed framework, and projected outcomes (model-based). Interpretation is the authors' reading.

12.1 Interpretation of Literature

The retained literature consistently identifies commuting as an addressable source of urban emissions and the reduction of private-vehicle reliance as the recurring strategy (Comi & Polimeni, 2024; Eslami, 2025). The benefit of each low-carbon option is conditional on infrastructure and urban design (Marzolla et al., 2025), on demographics, time, accessibility and affordability for electric micromobility (Wu et al., 2025), and on energy supply and uptake for electric vehicles (Ndhlovu et al., 2025). Digital tools are reported to support forecasting, route optimisation and personalised advice (Kashem et al., 2025; Ouyang et al., 2025), while behavioural determinants influence whether people act on information (Eslami, 2025). *Interpretation:* since benefits are conditional, no single mode suits every commuter, so a measurement design should record both psycho-social variables (awareness, concern, intention) and practical ones (cost, time, convenience), as this study does.

12.2 Findings from Simulated Data

In the simulated data, awareness and intention were positively associated ($\rho = .34$), convenience was associated with acceptance ($\rho = .21$), but concern was not significantly associated with mode shift ($\rho = .03$) and private-vehicle users did report lower intention. In the logistic model, cost sensitivity was a positive predictor and perceived convenience a negative predictor of a low-emission majority; psycho-social predictors added little once practical factors were included. These patterns were largely built into the generator (Section 8): the significant associations illustrate how they would be reported, the null results how non-significant ones would be reported, and none is evidence about real students. The baseline-versus-recommendation task produced a fall in private-car share from 35.0% to 25.6% and a modelled 18.7% reduction in CO_2 per trip, with a negligible time penalty (mean −0.15 min) and a modelled cost saving; the direction follows from the recommendation rule and the follow-probability model, so the magnitude depends on assumed inputs and not on observed behaviour.

12.3 Comparison with Previous Research

The simulated results cannot be compared with previous research as independent evidence, because agreement is by construction. Qualitatively, the design is consistent with the published emphasis on practical conditions (time, affordability, accessibility) for micromobility use (Wu et al., 2025) and on the combination of impact information with behavioural incentives (Eslami, 2025; Niazi et al., 2024). Whether real students respond similarly to carbon-aware recommendations is an empirical question that the retained sources do not answer. In particular, no retained source quantifies the effect of personalised carbon feedback on commuting mode choice, so the magnitude seen in the simulation must not be read as consistent or inconsistent with published effects.

12.4 Implications for System Design

For assistant design, the demonstration suggests that (i) personalisation by cost and time weights matters because practical factors dominated the baseline model; (ii) options with little time penalty and clear cost savings are the likeliest to be followed; and (iii) the recommendation rule should be transparent, because acceptance and trust were treated as attitudinal drivers of following suggestions. These are design hypotheses for testing, not validated findings. The MaaS literature also warns that digital integration alone does not guarantee environmental benefit unless low-carbon options are prioritised (An & Shen, 2025), which supports including CO_2 explicitly in the ranking.

12.5 Theoretical and Practical Implications

The paper contributes a measurement-to-system mapping: each research construct has a defined place in the five-layer architecture and a metric. Methodologically, it demonstrates reporting practice for simulated data (labelling, disclosure of built-in structure, caution on agreement with prior work). Practically, universities and developers can reuse the instrument and analysis plan for a genuine pilot, and can use the data dictionary and variable mapping to log the same indicators in a real system. The principal theoretical question left open is the relative contribution of psycho-social variables and practical constraints to real recommendation uptake. Infrastructure and policy remain preconditions: walking and cycling infrastructure, reliable public transport, real-time information and integrated ticketing enable the behaviours that assistants seek to encourage (Cui & Zhang, 2024; Marzolla et al., 2025).

13 Limitations

The limitations below are substantial and should be read before any result is interpreted.

- **Simulated rather than genuine data.** All numerical results come from a synthetic dataset with built-in structure; they describe no real person and cannot be cited as findings.
- **Proposed rather than completed sampling.** The survey, pilot, approvals and recruitment have not taken place; the sample size of 150 is a plan.
- **Convenience sampling (proposed)** would produce a non-probability sample with selection bias.
- **Single-institution context** limits generalisability to other universities, cities and age groups.
- **Self-report bias.** Awareness, concern and acceptance are self-reported and subject to social-desirability bias.
- **Scenario-based rather than real behaviour.** Stated choices in hypothetical trips may not predict actual mode choice, and the within-subject design cannot rule out demand effects.
- **Limited scenarios.** Six scenarios cannot represent all commuting conditions; two-wheelers, a common commuting mode, are not modelled as a scenario option.
- **Uncertain emission factors.** The factors are illustrative inputs; real values vary with fleet, occupancy, fuel and grid.
- **No real system deployment.** The commute assistant exists only as a design; no recommendation engine was built or tested.
- **No causal inference.** Cross-sectional associations and a within-subject hypothetical comparison cannot establish causal effects of recommendations.
- **Statistical limitations.** Several scales had modest reliability (time and convenience below .60), the outcome group size gave about 7.2 events per predictor, trip-level tests ignore clustering, and p-values reflect a generator, not a population.
- **Uncertain projections.** Monthly and annual projections rest on equal-weight scenario trips, a fixed trip count and an assumed persistence of effect.

14 Conclusion

The literature establishes that commuting is a significant, addressable source of urban emissions; that low-carbon options are conditional on infrastructure, energy supply and user circumstances; that AI, IoT and big-data tools can support forecasting, route optimisation and personalised advice; and that behavioural mechanisms are determinants of success. The main gap is the absence of integrated, personalised, behaviourally aware commute assistants and of measurable designs for evaluating them.

The quantitative methodology was designed to investigate sustainable-commuting awareness, concern, intention and mode choice, and the response to personalised recommendations. The paper specifies a cross-sectional, scenario-based design with research questions, hypotheses, variables, a seven-section instrument, a data dictionary, an emission model and a statistical analysis plan, all mapped to one another.

The simulated analysis demonstrated that the analysis plan runs end to end on a clearly labelled synthetic dataset of 120 respondents: descriptive statistics, reliability, normality checks, correlation, group comparisons, logistic regression, a conditional mode-choice model, a baseline-versus-recommendation comparison and a CO_2 projection. Of the six hypotheses, those built into the generator were detected and the others were not, which illustrates both significant and null reporting. In the simulated demonstration scenario, the model projected an illustrative 18.7% reduction in CO_2 per trip and an illustrative 5.4 kg saving per commuter per month under stated assumptions. These are demonstrations of the procedure and projections from a model; they are not evidence about real commuters or observed environmental effects.

The conceptual framework contributes a five-layer architecture with an explicit mapping from research variables to system inputs, AI processes, behavioural interventions and evaluation metrics. What remains unvalidated is everything empirical: real awareness, concern and intention levels; real recommendation acceptance and mode shift; real CO_2 savings; and the performance of any implemented assistant. These require genuine data collection and pilot deployment.

15 Recommendations

15.1 A. Supported by Existing Literature

1. Design commute assistants around multimodal options (public transport, active travel, micromobility, shared mobility and electric vehicles) rather than a single mode, because integrated ecosystems are under-researched and ease of multimodal planning is associated with acceptance (Kashem et al., 2025; Smith et al., 2023).
2. Base emission estimates on regional, regularly updated emission factors, because the benefit of electric options depends on the electricity source (Ndhlovu et al., 2025).
3. Orient MaaS and similar platforms explicitly towards low-carbon options, since adoption alone does not guarantee environmental benefit (An & Shen, 2025).
4. Combine environmental-impact information with behavioural mechanisms such as incentives rather than relying on information alone (Eslami, 2025; Niazi et al., 2024).
5. Pair digital tools with supportive infrastructure and policy, including walking and cycling infrastructure, reliable public transport, real-time transit information and integrated ticketing (Cui & Zhang, 2024; Marzolla et al., 2025).
6. Adopt interdisciplinary approaches that bring together urban planning, behavioural science and digital technology (Eslami, 2025).

15.2 B. Requiring Future Empirical Validation

1. Which nudging mechanisms (carbon feedback, gamified rewards, social comparison) are most effective, and how to weight CO_2, time, cost and health in the ranking, should be established by pilot evaluation, not assumed.
2. Claims about mode shift, recommendation acceptance and emission savings should be made only after the metrics in Section 11 have been measured with genuine data.
3. Whether psycho-social variables or practical constraints better predict real recommendation uptake should be tested with a genuine survey and observed behaviour.
4. The instrument's reliability and validity, in particular the short three-item scales, should be confirmed in a pilot before the main survey.

16 Future Scope

1. Real survey deployment at Chandigarh University after piloting and approvals, replacing the simulated dataset.

2. Larger samples (beyond $n \approx 150$) and multi-university samples to improve generalisability.
3. A pilot of the commute assistant with a small group of volunteers.
4. Integration of real mobility data and real transit data (schedules, delays, fares).
5. Validated regional emission factors, including two-wheelers and local grid mix.
6. A/B testing of recommendation designs (with and without CO_2 feedback, gamification, social comparison).
7. Longitudinal behavioural tracking of observed mode choice and engagement.
8. Pre-test/post-test intervention designs measuring changes in awareness and intention.
9. Real-world CO_2 measurement or logged-trip emission estimation.
10. Machine-learning optimisation and personalised recommendation algorithms trained on genuine data.
11. Comparison of recommendation algorithms (e.g., rule-based ranking versus learned policies).
12. Integration with MaaS platforms, designed to favour low-carbon options.
13. A more exhaustive systematic literature review to reduce selection bias inherent in the original narrative review.

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Note. Reference details are those of the original paper, preserved without addition. Entries for which the source material gave no venue (Mogire, 2025; Sunio, 2018) are left as given. All references listed here are cited in the text.

A Appendix A — Full Questionnaire

Participant questionnaire (proposed; not yet administered). *Informed consent:* You are invited to take part in a research study on sustainable commuting among university students. Participation is voluntary and you may withdraw at any stage. No names, passwords or financial details are collected. Responses are anonymous and used only for academic research. By proceeding you confirm that you are 18 or older and agree to participate.

Section A — Commuting profile

A1 Age (years). A2 Gender (Male / Female / Other-prefer not to say). A3 Year of study (1–4). A4 Programme (CSE / Other engineering / Management-Commerce / Sciences-Other). A5 One-way distance to campus (<2 km / 2–5 km / 5–10 km / >10 km). A6 Main current mode (Private car / Two-wheeler / Bus / Walking-Cycling / Carpool-Shared ride). A7 Days per week you travel to campus (3–6).

Section B — Sustainable mobility awareness (1 = strongly disagree ... 5 = strongly agree)

B1. I know which commuting options (e.g., walking, cycling, bus, metro, carpool, e-bike) produce lower CO_2 emissions than a private car. B2. I understand that private-car commuting contributes substantially to urban greenhouse-gas emissions. B3. I am aware that public transport can emit less CO_2 per passenger-km than a solo car trip. B4. I know that walking and cycling produce almost no tailpipe emissions and bring health benefits. B5. I am aware that carpooling or ride-sharing can reduce the number of vehicles on the road. B6. I am aware that shared e-bikes and e-scooters can replace short car trips. B7. I understand that the environmental benefit of electric vehicles depends on how the electricity is generated. B8. I know of mobile apps or digital tools that show multimodal routes or travel emissions. B9. I can roughly judge which of two travel options has the lower CO_2 impact. B10. Attention check (unscored): For quality control, please select "Agree" (4) for this item.

Section C — Environmental concern and attitudes (same 1–5 scale)

C1. I am concerned about air pollution caused by road traffic in my city. C2. I am concerned about the effect of transport emissions on climate change. C3. I feel a personal responsibility to reduce the emissions caused by my daily travel. C4. I feel uncomfortable choosing a high-emission option when a cleaner one is available. C5. I believe that individual commuting choices make a difference to the environment. C6. I think universities should encourage low-emission commuting. C7. Reducing my travel emissions is important to me even if it takes slightly longer.

Section D — Commuting decision factors (same 1–5 scale)

D1. Low travel cost is a major factor in my choice of commute mode. (cost1) D2. I would switch to a cheaper option even if it were less comfortable. (cost2) D3. I check the cost of alternatives before travelling. (cost3) D4. Short travel time is a major factor in my choice. (time1) D5. I would pay more to reach campus faster. (time2) D6. Unexpected delays strongly affect my choice of mode. (time3) D7. I prefer a door-to-door option that needs no transfers. (conv1) D8. Flexibility to leave at any time is important to me. (conv2) D9. Ease of planning and boarding (no long waits, no complicated tickets) is important to me. (conv3) D10. Comfort (seating, temperature, protection from weather) is important to me. (comfort) D11. Personal safety, including at night, is important to me. (safety) D12. Having a stop, station, bike or ride service close to my home and to campus is important to me. (access)

Section E — Sustainable travel intention

Over the next semester, how willing are you to use the following for your commute? (1 = very unwilling ... 5 = very willing). E1 Walking. E2 Cycling. E3 Public transport (bus/metro/rail). E4 Carpooling. E5 Shared mobility (e.g., shared e-scooter or pooled ride). E6 E-bike or e-scooter. E7 Electric vehicle (own or shared).

Section F — Scenario-based commute choice (Stage A)

For each of six trips (Appendix B) choose the option you would most likely take. Cards show time, cost, comfort and convenience.

Section G — Recommendation and assistant acceptance

G1: For each scenario, a personalised recommendation card (time, cost, CO_2, convenience, environmental benefit) is shown; choose again (Stage C). G2: Rate agreement (1–5) with: G2-1. Such an assistant would be useful for planning my commute. (acc1) G2-2. I would trust the time, cost and CO_2 information it shows. (acc2) G2-3. I would accept its recommended option when it is reasonable. (acc3) G2-4. I would be willing to follow its suggestions even if my usual mode is different. (acc4) G2-5. Seeing the CO_2 impact of each option would add value to my decisions. (acc5) G2-6. I would continue to use such an assistant over the semester. (acc6) G2-7. I would regularly check CO_2 feedback or my carbon-saving summary. (acc7)

B Appendix B — Scenario-Based Commute Choice Items

Scenario values are illustrative design values and do not describe real services. Comfort and convenience are 1–5 ratings shown to respondents as star ratings. CO_2 (g, from Equations 1–2) is displayed only in Stage B and is withheld in Stage A.

Table 26: Scenario option cards (time in minutes; cost in rupees).

Scenario	Option	Time	Cost	Comfort	Conv.	CO_2 (g)
S1: Short campus-area trip, fair weather, off-peak (3 km)	Private car	12	40	4	5	510
	City bus	20	10	2	3	210
	Bicycle	15	0	2	3	0
	E-bike	11	25	3	4	60
	Shared ride	13	30	3	4	180
S2: Peak-hour trip, congested corridor (6 km)	Private car	25	85	4	5	1020
	City bus	35	15	2	3	420
	Bicycle	28	0	2	3	0
	Carpool	27	30	3	3	342
	E-bike	19	45	3	4	120
S3: Hot, humid day, medium distance (8 km)	Private car	22	110	4	5	1360
	City bus	38	12	2	3	560
	Metro/rail	28	25	4	4	240
	Carpool	25	40	3	3	456
	Shared ride	30	70	3	4	480
S4: Rainy day, longer distance (12 km)	Private car	35	170	4	5	2040
	City bus	50	15	2	3	840
	Metro/rail	38	40	4	4	360
	Carpool	38	60	3	3	684
	Shared ride	40	110	3	4	720
S5: Long trip with first/last-mile leg (15 km)	Private car	40	210	4	5	2550
	Metro + e-bike	45	45	4	3	420
	City bus	60	30	2	3	1050
	Carpool	44	75	3	3	855

Scenario	Option	Time	Cost	Comfort	Conv.	CO_2 (g)
	Shared ride	48	140	3	4	900
S6: Evening return trip, personal-safety concern (5 km)	Private car	18	70	4	5	850
	City bus	30	15	2	3	350
	Carpool	20	30	3	3	285
	Shared ride	22	60	3	4	300
	E-bike	15	35	3	4	100

C Appendix C — Variable Coding Scheme

Table 27: Coding and scoring rules.

Item	Rule
Likert items	1 = strongly disagree / very unwilling … 5 = strongly agree / very willing; no reverse-keyed items
Scale scores	Arithmetic mean of items (awareness: aw1–9; concern: ec1–7; cost: cost1–3; time: time1–3; convenience: conv1–3; intention: int1–7; acceptance: acc1–acc6)
`engagement`	Mean of acc6 and acc7
Mode codes	C private car; B city bus; M metro/rail; Y bicycle; E e-bike; P carpool; S shared ride; X metro + e-bike
Sustainable choice	Any code other than C
`low_emission_majority`	1 if at least 4 of 6 Stage A choices are not C
Recommendation followed	Stage C = recommendation and recommendation ≠ Stage A
Acceptance rate	Followed ÷ number of decisions where recommendation ≠ Stage A
Mode shift	Stage A = C and Stage C ≠ C; `shift_count` = number of such scenarios; indicator = 1 if count ≥ 1
Emission per decision	$E = d \times EF_m$ (Section 7); `baseline_emission` = mean over six Stage A choices; `recommended_emission` = mean over six Stage C choices
Attention check / exclusion	Fail attention item, speeding (< one-third median time) or straight-lining → excluded (proposed rule)

D Appendix D — Item-Level Data Dictionary

Item-level variables and their meanings are listed in Table 6 (Section 6) and in the full questionnaire (Appendix A). Each item is an integer from 1 to 5. Items map to constructs as follows: aw1–aw9 → `awareness_score`; ec1–ec7 → `concern_score`; cost1–cost3 → `cost_sensitivity`; time1–time3 → `time_sensitivity`; conv1–conv3 → `convenience_score`; int1–int7 → `intention_score`; acc1–acc6 → `acceptance_score`; acc6–acc7 → `engagement`. The complete simulated dataset, including all item-level values, accompanies this paper as `simulated_dataset.csv`.

E Appendix E — Simulated Dataset

The full synthetic dataset of $N = 120$ respondents is provided as a machine-readable file (`simulated_dataset.csv`) generated by `simulation_analysis.py` with fixed seed `20251006`. Table 28 reproduces a representative extract of 20 respondents. The extract is sufficient to show the structure; the full file contains every item and every derived variable described in Section 6. **These are simulated values: the respondents are fictional and no real person provided any value.**

Table 28: Representative extract of the simulated demonstration dataset (first 20 of 120 fictional respondents).

ID	Age	G.	Dist.	Current mode	Aware.	Intent.	Accept.	CO_2 saved (g)
R001	21	F	<2	Carpool/Shared	4.44	4.71	4.33	57
R002	20	F	<2	Walking/Cycling	4.56	3.86	3.83	0
R003	18	F	2–5	Bus	3.89	3.14	4.33	187
R004	19	M	>10	Bus	3.78	3.14	4.00	74
R005	21	M	>10	Carpool/Shared	3.33	3.57	3.67	150
R006	19	M	5–10	Private car	1.89	4.00	3.17	320
R007	19	F	>10	Bus	4.00	2.86	3.83	105
R008	20	F	>10	Carpool/Shared	2.56	3.00	2.83	60
R009	18	M	2–5	Walking/Cycling	3.11	3.43	3.67	105
R010	21	M	>10	Bus	3.67	4.00	4.17	53
R011	22	F	5–10	Private car	3.11	4.29	1.83	0
R012	20	F	5–10	Bus	2.67	2.29	3.50	0
R013	21	F	5–10	Carpool/Shared	2.22	3.00	4.33	159
R014	20	M	<2	Private car	4.22	3.86	5.00	537
R015	20	M	2–5	Bus	3.11	4.14	4.17	266
R016	18	M	<2	Bus	4.78	4.71	3.67	54
R017	21	M	2–5	Private car	3.00	3.43	3.50	0
R018	21	M	>10	Two-wheeler	4.67	4.43	3.33	99
R019	20	M	<2	Walking/Cycling	4.44	4.00	3.33	85
R020	19	M	>10	Carpool/Shared	3.11	4.14	4.00	592

Generation summary. Demographics and commute-distance categories were drawn from the assumed distributions in Section 8; current mode was drawn conditional on distance; correlated latent traits were transformed into Likert items with added noise; Stage A choices came from a random-utility model; recommendations were produced by a personalised weighted ranking; and Stage C choices followed the recommendation with a probability that increased with acceptance. All steps used the single seed `20251006`, so the dataset and every statistic in Section 9 are exactly reproducible.

F Appendix F — Statistical Analysis Output Tables

The complete machine-readable statistical output is provided as `analysis_output.json`. It contains: descriptives (mean, SD, median, min, max, skewness, kurtosis, Shapiro–Wilk) for every composite; Cronbach's alpha for every multi-item scale; the full Spearman correlation matrix; the group-comparison statistics (Welch t with degrees of freedom and Cohen's d; Mann–Whitney U with rank-biserial r; Kruskal–Wallis H); the binary logistic-regression coefficients, standard errors, p-values, odds ratios and confidence intervals; the conditional-logit attribute coefficients; the baseline-versus-recommendation mode counts and indicators; the per-scenario CO_2 means; the projection figures; and the hypothesis-testing summary with Holm-adjusted p-values. The key values are reproduced in Tables 12–23 of the main text.

G Appendix G — Five-Layer System Architecture

The five layers are summarised in Table 29. Each layer is conceptual and has not been implemented.

Table 29: Five-layer architecture (conceptual) and its research linkage.

Layer	Function	Research linkage
1 — User input	Origin, destination, time, preferences, constraints	`commute_distance`, `current_mode`, `commute_days`, decision-factor scores
2 — Data acquisition	Traffic, transit, weather, air quality, emission factors	Layer 2 feeds the emission model (Section 7); factors would be regional
3 — AI processing	Route generation, emission estimation, multimodal optimisation, prediction	Scenario option cards; Equations 1–3
4 — Decision and nudging	Multi-criteria ranking, carbon feedback, gamification, social comparison	Stage B recommendation card; personalised weights on time, cost, CO_2
5 — Feedback and evaluation	Log selected mode, measure impact, feed back to personalise	Stage C choices; mode-shift, acceptance, CO_2 and engagement metrics

H Appendix H — Proposed Evaluation Metrics

The proposed evaluation metrics, their definitions and their measurability are given in Table 25. In the present paper only the metrics marked "Simulated" were computed, and only on the simulated dataset. The remaining metrics (awareness, concern and intention change) require a genuine pre/post design and were not computed.